

M.Sc Life Science Informatics - Data Visualization

The effect of different parameters on sleep quality

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1. Introduction

One of the many definitions of sleep, define it as an everchanging circadian-regulated state between asleep and awake. A complete cycle of sleep is characterized by three cycles of non-rapid eye movement sleep (N_1, N_2, N_3) followed by rapid eye movement sleep. Sleep quality is optimal when consisting of 4 to 5 of such sleep cycles. Suppression of signals and arousal systems which keep the organism in a sleep state, lowers with the increase of extracellular adenosine which activates a specific area of the brain and acts as a state-switch.

The circadian rhythm, also referred to as the circadian clock, is influenced by internal signaling pathways and outside factors called “zeitgebers”. The main zeitgeber is light exposure, in lack of which, the pineal gland in the brain begins melatonin production which affects sleep regulation. The associated brain region with circadian clock activity is known as the suprachiasmic nuclei which is located within the hypothalamus (Baranwal et al., 2023).

1.1 Physiological impacts of impaired sleep

Patients with cardiovascular diseases are shown to have disrupted sleep, reduced REM time and slow wave sleep, fatigue, excessive sleep, shortness of breath at night or paroxysmal nocturnal dyspnea.

Sleep quality and duration, also, impacts cardiovascular factors and disease, the latter of which is one of leading causes of death. The inclusion of the measurement of sleep diseases such as sleep apnea and perturbations of insufficient or excessive sleep duration, helped improve the prediction of cardiovascular diseases. Shortened sleep duration leads to hormonal disbalances such as: cortisol and testosterone decrease, imbalance of ghrelin and leptin which increase appetite (Baranwal et al., 2023), catecholamine increase, upregulated levels of interleukins and inflammatory cytokines such as tumor necrosis factor-α, plasma norepinephrine increase, the latter of which leads to insomnia (positive feedback loop), fragmented sleep, obesity, insulin resistance, and many more. Other factors that were detected were a rise in heart rate but decrease in its variability, and heightened blood pressure (Jaspan et al., 2024).

The restriction of the upper airway during sleep, restricting aeration levels from low to none, are the defining feature of sleep apnea and hypopnea. Sleep apnea, due to hypoxia conditions, can cause an increase in ROS and inflammation which can eventually lead to symptoms such as hypertension, arrhythmias, metabolic syndrome, or even heart failure.

Inflammation is considered the main linking factor between cardiovascular diseases and sleep quality, since many immunological processes that participate in homeostasis happen during sleep circadian cycles. Venous and arterial function is

obstructed after several days of poor sleep. Issues arise due to predominance of the parasympathetic system due to sleep imbalance.

Sleep health varies and changes according to an individual's sex, income, environmental exposure to pollutants, substance abuse; variances have been observed that are linked to race and ethnicity as well.

The association between cardiovascular disease and sleep health is proven but there is not much data regarding the improvement of sleep in prevention of cardiovascular diseases (Jaspan et al., 2024).

2. Problem Definition

Research question: How do different parameters influence sleep quality?

Hypothesis: The effect of different parameters on sleep quality.

Sleep is fundamental for a wide range of cognitive, emotional, and behavioral functions, such as attention, memory, executive abilities, emotional regulation, and social interactions. Disruptions in sleep can negatively affect these processes and therefore lead to noticeable decline in everyday functioning, work performance and in some instances, safety. Obtaining enough sleep helps to maintain alertness, focus, and cognitive efficiency, while insufficient sleep is linked to reduced attention, weaker cognitive control, and impaired processing of sensory information (Hyndych et al., 2025).

Sleep insufficiency represents an important public health concern, with a significant proportion of the population experiencing inadequate sleep. Tompa (2025) reports that more than one third of adults in the U.S. and nearly 80% of adolescents do not obtain sufficient sleep. Furthermore, approximately one quarter of adults experience chronic sleep disorders, including conditions such as insomnia and sleep apnea.

In a different population, Oros et al. (2025) observed that individuals' perceptions of their sleep quality are affected by the combined effects of lifestyle habits, environmental conditions, and health-related factors.

All this gathered information led to focus this study’s aim to question which particular parameters affect sleep quality and how much influence they provide.

3. Objectives

The aim of this project is to observe how people’s health status and other various aspects of daily life such as stress, occupation, physical activity levels, etc., influence their quality of sleep and potential predisposition to sleep diseases such as insomnia and sleep apnea. The focus is to detect a pattern in which some or many aspects of life determine the quality of sleep in patients.

4. Methodology

4.1. Study Design and Data Source

This project was a collaborative data analysis study in which the association of behavioral and health-related parameters and sleep quality was investigated. This analysis included multiple statistical and machine learning approaches to explore the relationship between some parameters and sleep quality. Each member was responsible for different analytical approaches.

The dataset used for this study was obtained from Kaggle under the name of "Sleep Health and Lifestyle Dataset" published by Laksika Tharmalingam (Tharmalingam, 2023). It is a synthetic dataset created by the publisher that contains 374 individuals, and contains various variables related to sleep and daily habits such as: sleep metrics, lifestyle factors, cardiovascular health and sleep disorder occurrence.

4.2. Data Cleaning

Before the statistical analysis, the data was observed for variable types, data completeness and consistency. Variables were categorized into continuous and categorical variables to determine the appropriate statistical methods used.

In order to be able to work with some variables it was necessary to modify some. The Blood Pressure variable was modified from systolic and diastolic reading to a categorical variable where depending on the reading it was assigned to a "Normal", "Elevated", and "High" category (Mayo Clinic, n.d).

4.3. Analysis

4.3.1 Descriptive Statistics

Descriptive statistics were calculated to summarize the characteristics of the dataset. Continuous variables were described using measures of central tendency and dispersion, including mean, median, standard deviation, minimum and maximum values. Categorical variables were summarised using absolute counts and percentages. To visualize the distribution of variables, histograms and bar charts were created.

4.3.2 Decision Trees

Decision Trees were used as a simple supervised parametric method to try to determine which features are most associated with a certain outcome, in this instance whether certain features lead to sleep disorder or a difference in sleep quality. Importance was not given to feature importance.

Sleep quality is a column representing a range where 1 is the worst quality sleep and 10 is the best. Since we are attempting to classify the instances and predict feature association, this column was separated into three categories: low quality with values 1-4, medium quality with values 5-7, and high quality with values until and including 10. Sleep disorder was the other feature that was used as an outcome instead.

The following libraries were created to create a random partition of training and test data, to plot the decision tree, and subsequently, create the confusion matrix: rpart, rpart.plot, and caret. No hyperparameter tuning was applied to the model.

4.3.3 k-means clustering

K-means clustering algorithm was used as an unsupervised method without any outcomes to see the potential existence of natural clusters in the groups and if such an existence is confirmed, what are the characterizing features.

Non-numeric features were turned into numeric and the dataset was standardized as it is required for clustering algorithms. The “factoextra” library was used to calculate the optimal number of clusters as no hyperparameter tuning was done. The number of clusters with the last detrimental lowest total within the sum of squares was selected for the method.

Subsequently, a number-based instance-distribution table and a percentage-based instance-distribution table were added to show the dispersion of the classes of sleep disorder and sleep quality in the clusters. The averages of each feature for each cluster were calculated and then compared to the averages of each feature of the dataset. The comparison was done in order to attempt to infer defining qualities about each cluster.

4.3.4 Non-Parametric Statistical Tests

Due to the dataset containing a mixture of ordinal and categorical predictors, a fully parametric approach would have led to check separately many assumptions. Therefore, a non-parametric approach was selected for this study. Another reasoning for this decision was that the dependent variable, sleep quality, was measured on an ordinal scale (1 to 10), and several predictor variables were categorical.

The association between continuous variables and sleep quality was assessed using Spearman's rank correlation. The correlation coefficient was used to describe the direction and strength of the association.

For comparisons of two independent groups, the Mann–Whitney U test was applied. This test was used to determine whether sleep quality differed significantly between two categories, such as in the case of Gender. For variables that contained more than two independent groups, the Kruskal-Wallis test was applied. This test was

used to evaluate whether statistically significant differences in sleep quality existed between multiple categories, such as occupation groups.

All statistical tests were performed using a two-sided significance level of 0.01. A p-value below 0.01 was considered statistically significant.

To support the interpretation of the statistical tests, graphical visualizations were created. Correlation results were visualized to demonstrate the strength and direction of associations between variables. Additionally, p-values from the statistical tests were summarized using a heatmap to provide an overview of significant and non-significant relationships across variables. To improve the visual interpretation, p-values were transformed using the negative base-10 logarithm. This transformation increases the visual separation between statistical significant and non-significant results.

4.3.5 Linear Regression

Multiple linear regression was used to model the relationship between sleep quality (dependent variable) and the available demographic, lifestyle, and health-related predictors. The objective was to identify variables associated with sleep quality while simultaneously accounting for the effects of the remaining predictors.

Prior to model fitting, categorical variables were converted to factors to ensure they were treated appropriately by the regression model. A full multiple linear regression model containing all available predictors was initially fitted using the lm() function in R.

The regression coefficients were estimated using the ordinary least squares (OLS) method by minimizing the Residual Sum of Squares (RSS), resulting in the regression model that best fit the observed data.

Model performance was evaluated using the adjusted R-squared, which measures the proportion of variability in sleep quality explained by the predictors while accounting for model complexity. The overall significance of the regression model and the individual predictors were assessed using the F-test and t-tests, respectively.

To evaluate the generalization performance of the regression model, the dataset was divided into training (80%) and test (20%) sets using stratified sampling based on blood pressure categories. The training set was used to fit the regression models and perform feature selection, while the test set was reserved for evaluating predictive performance on previously unseen observations.

4.3.6 Feature Selection

To obtain a more parsimonious model and reduce the model complexity, backward stepwise selection based on the Bayesian Information Criterion (BIC) was applied. The procedure started with the full regression model containing all predictors and

iteratively removed variables that did not improve the BIC. The final model was selected as the model with the lowest BIC, balancing model fit and complexity.

4.3.7 Model Evaluation

To verify that the assumptions of linear regression were reasonably satisfied, standard diagnostic analyses were performed. Residual plots were examined to assess linearity and homoscedasticity, Normal Q–Q plots were used to evaluate the normality of the residuals, and Residuals versus Leverage plots with Cook's distance were inspected to identify potentially influential observations. Multicollinearity among the predictors was assessed using the Generalized Variance Inflation Factor (GVIF). For categorical predictors with multiple levels, adjusted GVIF values were interpreted, where values below 5 were considered indicative of acceptable multicollinearity.

4.3.8 Model Diagnostics

To verify that the assumptions of linear regression were satisfied, diagnostic analyses were performed. Residual plots were examined to assess linearity and homoscedasticity, Normal Q–Q plots were used to evaluate the normality of residuals, and Cook's distance was inspected to identify influential observations. Multicollinearity among predictors was assessed using the Generalized Variance Inflation Factor (GVIF), where adjusted GVIF values were interpreted for categorical predictors with multiple levels.

4.4. Software

All analyses were conducted using the R program in RStudio. Descriptive statistics, statistical testing, and visualizations were performed using the following R libraries: dplyr, effectsize, rstatix, ggplot2, ggcorrplot, caret, car, kableExtra, and others.

5. Results

5.1. Descriptive Statistics

This study included a total of 374 patients. Descriptive statistics were performed to summarize the distributions of all variables before conducting the further analyses.

For the purpose of this study course, only the distribution of the most relevant variables were visualized.

5.1.1 Sleep Quality

The distribution of sleep quality scores are shown in Figure 1. Sleep Quality was measured on a scale from 1 to 10. Most participants reported a relatively high sleep quality, with scores concentrated between 6 and 9, being 6 and 8 the most frequent ratings. No participants scored below 4.

Overall, the distribution indicates that the majority of the patients perceived their sleep quality from moderate to good.

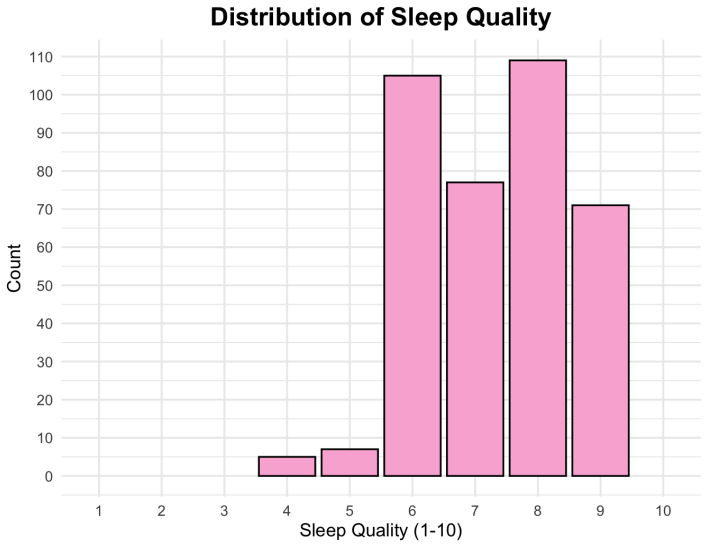


Figure 1. Distribution of the population's Sleep Quality on a scale from 1 to 10.

5.1.2 Sleep Duration

Figure 2 provides de visualization of the distribution of Sleep Duration. This variable ranged from approximately 5.5 to 8.5 hours, with the largest proportion of individuals sleeping between 7.5 to 8.5 hours per night. This distribution suggests that most of the individuals achieved sleep durations close to the recommended seven to eight hours per night.

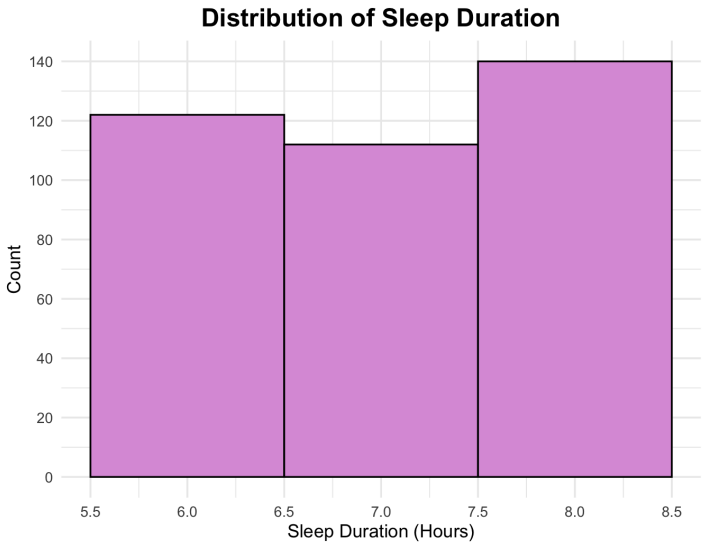


Figure 2. Population's Sleep Duration in Hours.

5.1.3 Occupation

The distribution of occupations is presented in Figure 3. The largest occupational groups being Nurses, Doctors, and Engineers, followed by Lawyers, Teachers, and Accountants. Occupations such as Managers, Sales Representatives, Software Engineers, and Scientists were underrepresented.

The unequal distribution across occupations should be considered when interpreting comparisons among occupation groups.

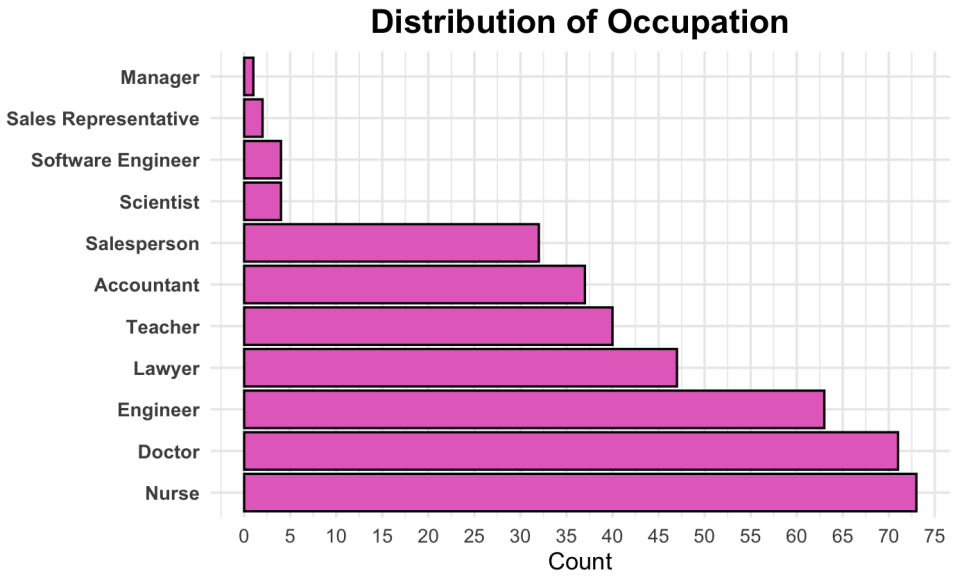


Figure 3. Distribution of Occupation types from the population.

5.1.4 Sleep Disorder

Figure 4 summarizes the prevalence of sleep disorders within the dataset. Most participants reported not having a diagnosed sleep disorder. Among patients with disorders, only two were reported. Both Sleep Apnea and Insomnia occur with similar frequency substantially with a smaller proportion of the sample compared to those without a sleep disorder.

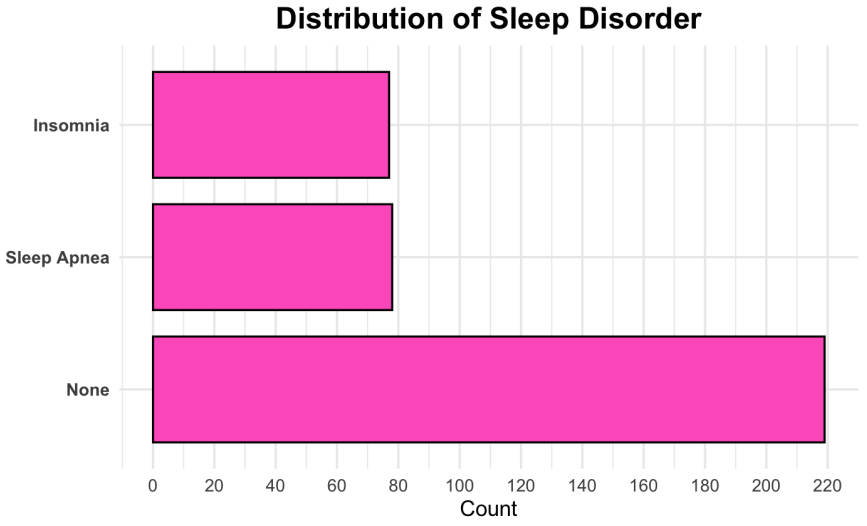


Figure 4. Sleep Disorder Prevalence in the population.

5.2. Decision Trees

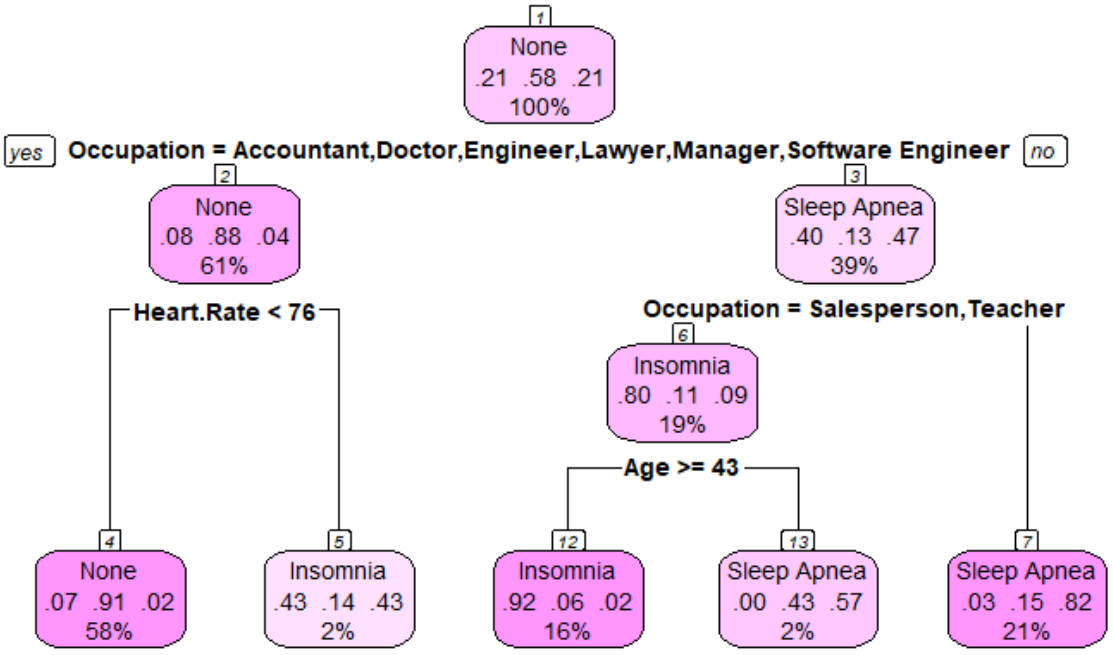


Figure 5. Decision tree feature split in regards to sleep disorder outcome

```
Confusion Matrix and Statistics

             Reference
Prediction    Insomnia None Sleep Apnea
  Insomnia          14    0           2
  None               1   41           1
  Sleep Apnea        0    2          12

Overall Statistics

               Accuracy : 0.9178
                 95% CI : (0.8296, 0.9692)
    No Information Rate : 0.589
    P-Value [Acc > NIR] : 3.722e-10

                  Kappa : 0.8554

 Mcnemar's Test P-Value : 0.343

Statistics by Class:

                     Class: Insomnia Class: None Class: Sleep Apnea
Sensitivity                   0.9333      0.9535              0.8000
Specificity                   0.9655      0.9333              0.9655
Pos Pred Value                0.8750      0.9535              0.8571
Neg Pred Value                0.9825      0.9333              0.9492
Prevalence                    0.2055      0.5890              0.2055
Detection Rate                0.1918      0.5616              0.1644
Detection Prevalence          0.2192      0.5890              0.1918
Balanced Accuracy             0.9494      0.9434              0.8828
```

Figure 6. Confusion matrix prediction measurements regarding sleep disorder decision tree

Important features in the class prediction for sleep disorder proved to be: Occupation, followed by Age and Heart Rate. 58% of observations which were in the

specified occupation and have a lower heart rate than 76 had an almost pure node of no sleeping disorders. In the other leaf, a heart rate above this threshold, had 2% of the test data set with associated insomnia.

The rest of the 40% of observations spread in the other node, with the purest node being that of observations with occupation of sales representative or teacher and age bigger or equal to 43. Sixteen percent of total observations pooled here with a high probability of insomnia. All other leaves in this partition, were classified into sleep apnea although there are wrongfully-classified observations.

The model has an accuracy of 91% having classified the majority of the binary class instances into their correct respective classifier.

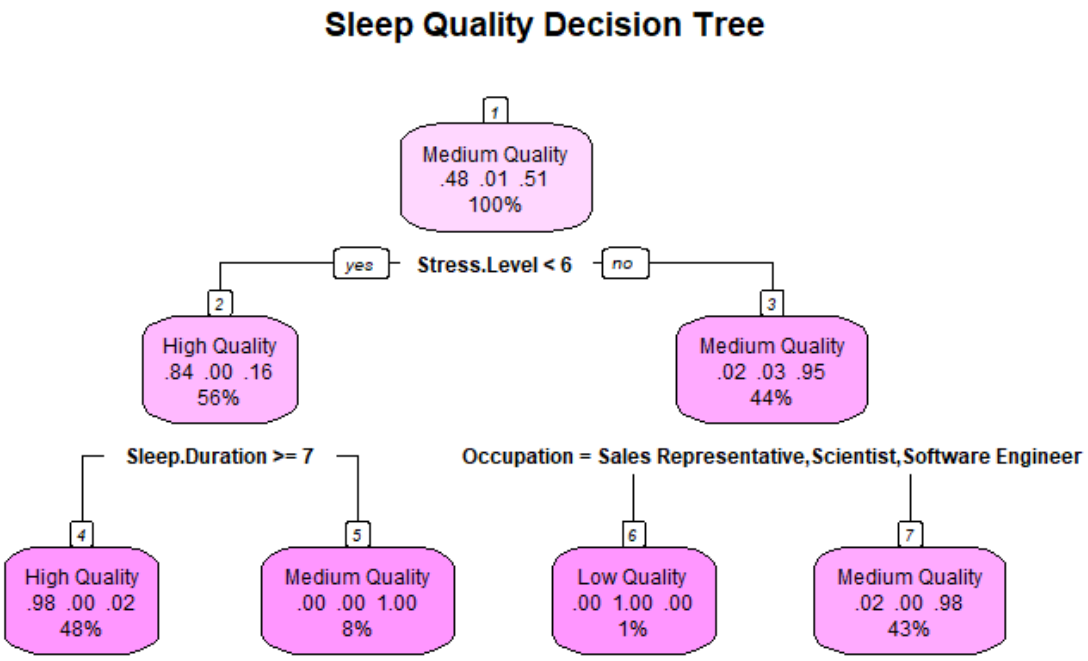


Figure 7. Decision tree feature split in regards to sleep quality outcome

The sleep quality decision tree classifier turned out more accurate than the sleep disorder classifier, with an accuracy of 97% and with two pure nodes and two almost pure nodes.

Reported stress quality of equal or lesser than six greedily separated the entirety of the high quality sleep instances, showing that in this dataset, this feature was key in the prediction of sleep quality. High quality sleep nodes were further purified by a node of sleep duration equal to or greater than seven hours.

```
Confusion Matrix and Statistics

                Reference
Prediction       High Quality Low Quality Medium Quality
  High Quality             36           0              2
  Low Quality               0           1              0
  Medium Quality            0           0             35

Overall Statistics

               Accuracy : 0.973
                 95% CI : (0.9058, 0.9967)
    No Information Rate : 0.5
    P-Value [Acc > NIR] : < 2.2e-16

                  Kappa : 0.9474

 Mcnemar's Test P-Value : NA

Statistics by Class:

                     Class: High Quality Class: Low Quality
Sensitivity                       1.0000            1.00000
Specificity                       0.9474            1.00000
Pos Pred Value                    0.9474            1.00000
Neg Pred Value                    1.0000            1.00000
Prevalence                        0.4865            0.01351
Detection Rate                    0.4865            0.01351
Detection Prevalence              0.5135            0.01351
Balanced Accuracy                 0.9737            1.00000
                     Class: Medium Quality
Sensitivity                         0.9459
Specificity                         1.0000
Pos Pred Value                      1.0000
Neg Pred Value                      0.9487
Prevalence                          0.5000
Detection Rate                      0.4730
Detection Prevalence                0.4730
Balanced Accuracy                   0.9730
```

Figure 8. Confusion matrix prediction measurements regarding sleep disorder decision tree

The first split is almost equal, with almost all instances of medium and low quality observations being split to the right. The positive features of any of the following occupations: sales representative, scientist, or software engineer, resulted in a small sample of low sleep quality observations.

Classification of true positives for all observations, no matter the quality, was entirely accurate with the slight exception of medium quality. Similar case for true negatives with the exception of a small portion of high sleep quality observations.

To summarize, the sleep quality decision tree classifier predicted that stress level and sleep duration were the most important features for high quality sleep. Occupation turned out to be a strong feature in prediction of sleep disorders.

5.3 k-means clustering

Due to the randomization and re-prediction of the model in each run, the value of the optimal number of clusters moved between the ranges of 5 to 7. As seen in the graph below, the chosen value is the one where the total within the sum of squares is the lowest before small impactless TSS differences.

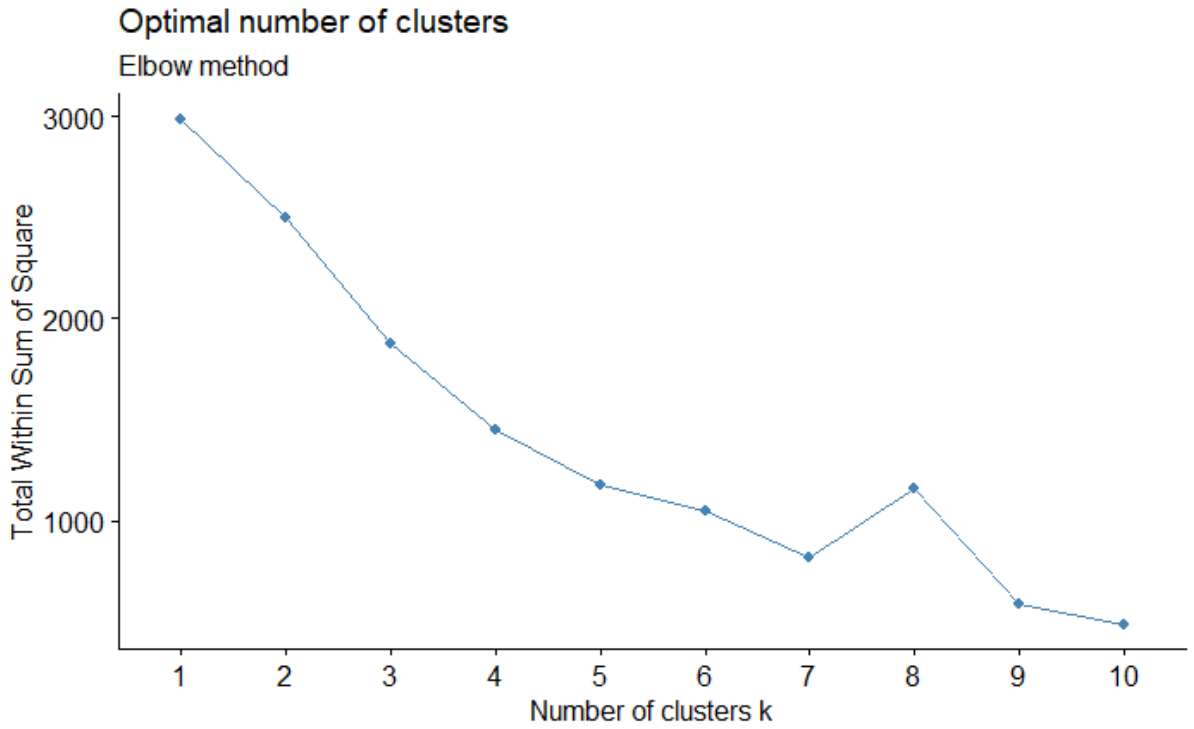


Figure 9. Elbow plot for computing the optimal number of clusters graphed against the total within the sum of squares.

The tables above the graph help us see the distribution of outcome classes throughout the clusters.

Table 1. Number-based instance-distribution table and a percentage-based instance-distribution of dataset observations across clusters, based on sleep disorder

clusters	Insomnia	None	S. Apnea	Insomnia	None	S. Apnea
1	1	2	29	0.031	0.062	0.9
2	1	34	30	0.015	0.52	0.46
3	1	31	1	0.03	0.93	0.03
4	4	100	5	0.036	0.91	0.04
5	7	34	0	0.17	0.82	0
6	25	7	1	0.75	0.21	0.303
7	38	11	12	0.62	0.18	0.196

Table 2. Number-based instance-distribution table and a percentage-based instance-distribution of dataset observations across clusters, based on sleep quality

clusters	High Q.	Low Q.	Medium Q.	High Q.	Low Q.	Medium Q.
1	0	0	32	0	0	1
2	65	0	0	1	0	0
3	0	0	33	0	0	1
4	75	0	34	0.68	0	0.31
5	34	0	7	0.82	0	0.17
6	6	0	27	0.18	0	0.81
7	0	5	56	0	0.08	0.91

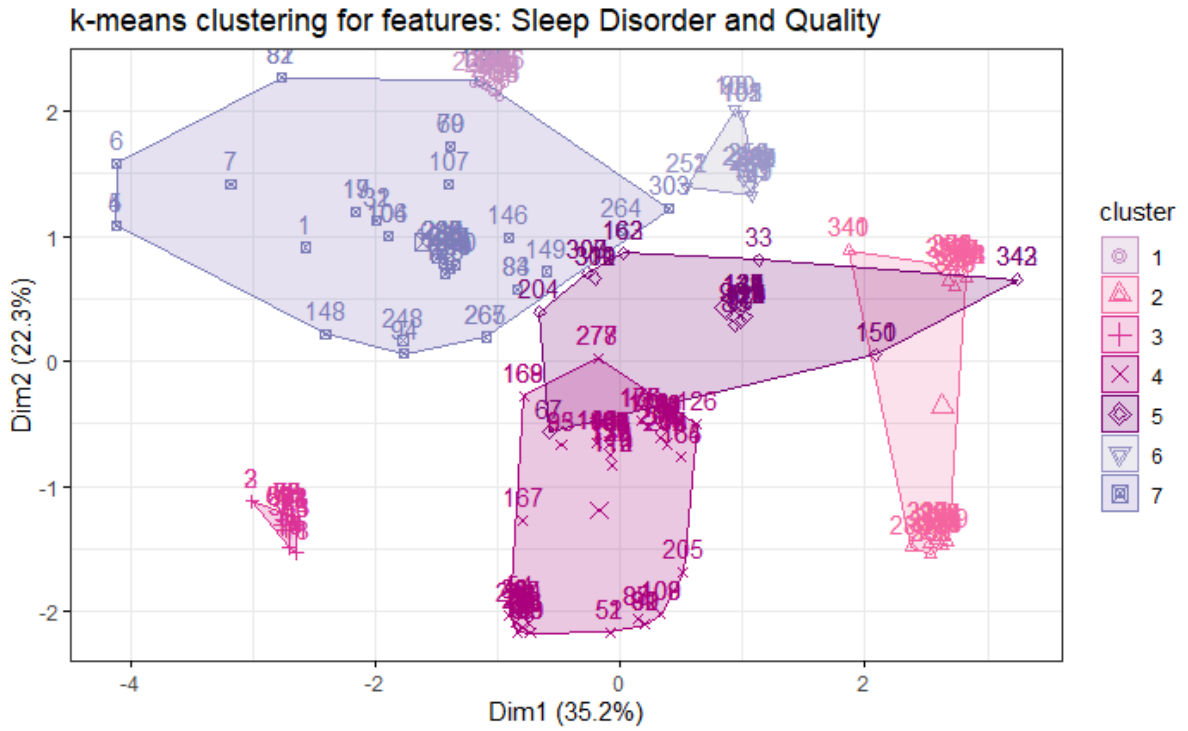


Figure 10. Cluster plot without the desired outcome variables: sleep disorder and quality.

Table 3. Calculated averages of each numerical features of the clusters

clusters	Age	Sleep Duration	Physical activity level	Stress level	Heart rate
1	38.2	7.28	62.34	4.14	68.11
2	39.7	6.46	44.13	6.76	75.18
3	37.08	7.5	73.99	5.15	69.52
4	42.84	6.7	46.81	4.12	65.54
5	31.55	6.06	31.81	8	71.81
6	55.46	8.25	52.84	3.03	66.64

7	50.12	6.16	82.17	7.79	74.46

Table 4. Calculated averages of each numerical features of the whole dataset

	Age	**Sleep Duration**	**Physical activity level**	**Stress level**	**Heart rate**
df mean	42.18	7.13	59.17	5.38	70.16

Comparison of the cluster feature average against the dataset average helps us deduce information about the cluster. For example, a lower than dataset average of feature age for cluster 1 tells us that the cluster is filled with younger people. Tables 1 and 2 help us associate sleep disorders and sleep quality with predominant features of a cluster.

5.4. Non-Parametric Statistical Tests

Due to several variables being ordinal or not satisfying the assumptions required for parametric testing, non-parametric statistical methods were used to evaluate associations with sleep quality.

5.4.1 Spearman Rank Correlation

Spearman's rank correlation was performed to evaluate relationships between continuous or ordinal variables and sleep quality.

As shown in Figure 11, Sleep Duration demonstrated the strongest positive correlation with sleep quality (p = 0.89), indicating that longer sleep duration was highly associated with a higher sleep quality.

On the other end, Stress Level showed the strongest negative correlation (p = -0.91), suggesting that increased stress was associated with lower sleep quality. Other strong negative correlation was observed for Heart Rate (p = -0.74). Age demonstrated a moderate positive correlation (p = 0.42), while Physical Activity and Daily Steps showed weak positive correlations.

These findings indicate that Stress Level and Sleep Duration were the variables with the strongest association with Sleep Quality.

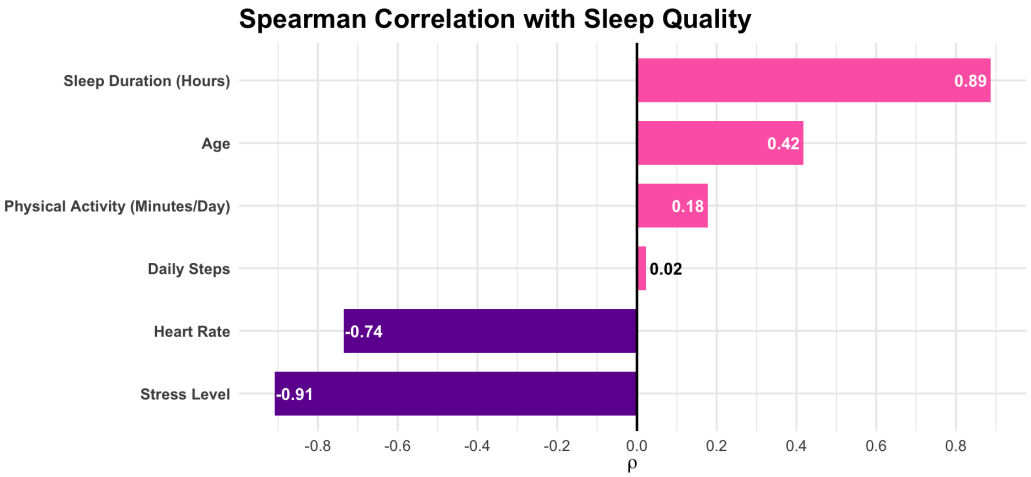


Figure 11. Spearman positive and negative correlations with Sleep Quality.

5.4.2 Mann-Whitney U Test

The Mann-Whitney U Test was used to compare Sleep Quality between categorical variables with only two groups, which in this case was only the Gender variable.

Figure 12 compares sleep quality between female and male individuals from our dataset. Female individuals reported slightly higher Sleep Quality scores than male participants. The difference was statistically significant (p-value = 6.05×10^{-9}). The effect size was r = 0.34, indicating a moderate effect, which suggests that gender accounted for a meaningful difference in reported Sleep Quality.

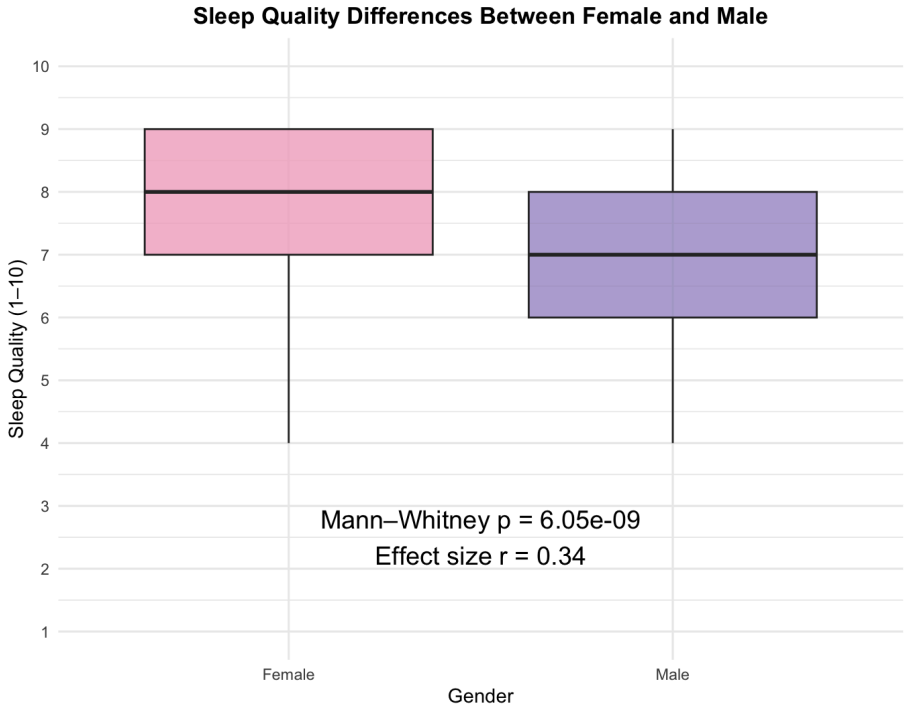


Figure 12. Box plot showing the Sleep Quality Differences Between Genders and its significance.

5.4.3 Kruskal-Wallis Test

The Kruskal-Wallis Test was performed to compare Sleep Quality across categorical variables containing more than two groups.

The largest effect size was observed for Occupations, with an effect site of $\varepsilon^2 = 0.43$, representing a large effect. This indicates that there are substantial differences in sleep quality among occupational groups. Sleep Disorder showed a medium effect ($\varepsilon^2 = 0.13$), suggesting that sleep quality differed between individuals with different sleep disorder statuses.

In contrast, Blood Pressure demonstrated a small effect ($\varepsilon^2 = 0.04$), indicating relatively minor differences in sleep quality across blood pressure categories.

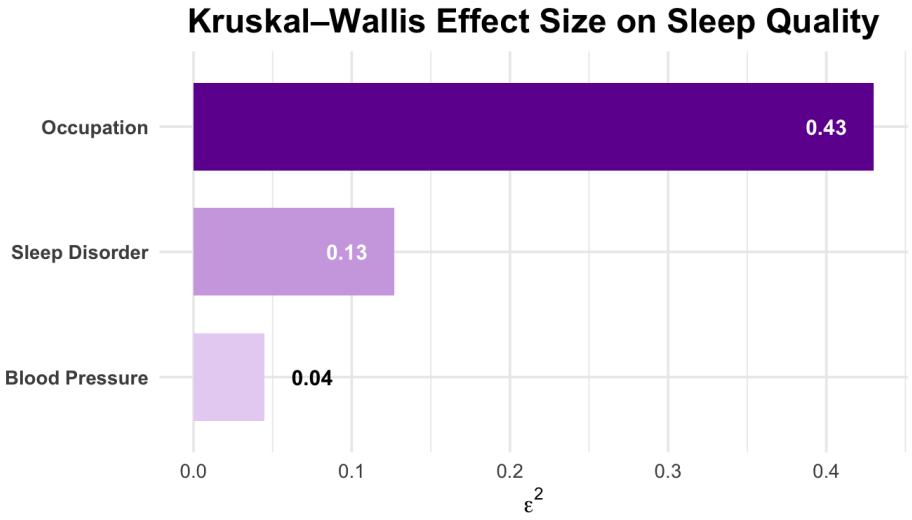


Figure 13. The effect size of variables with more than 2 groups on Sleep Quality .

5.4.4 Overall Statistical Significance

To facilitate comparison across all statistical tests, the p-values were transformed using $-\log_{10}(p)$ (Figure 14). Larger values indicate stronger statistical evidence against the null hypothesis.

Among all variables studied, Stress Level exhibited the strongest statistical significance, followed by Sleep Duration and Heart Rate. Significant associations were also observed in Occupation, Age, Sleep Disorder, Gender, Blood Pressure and Physical Activity. On the contrary, Daily Steps demonstrated minimal statistical significance.

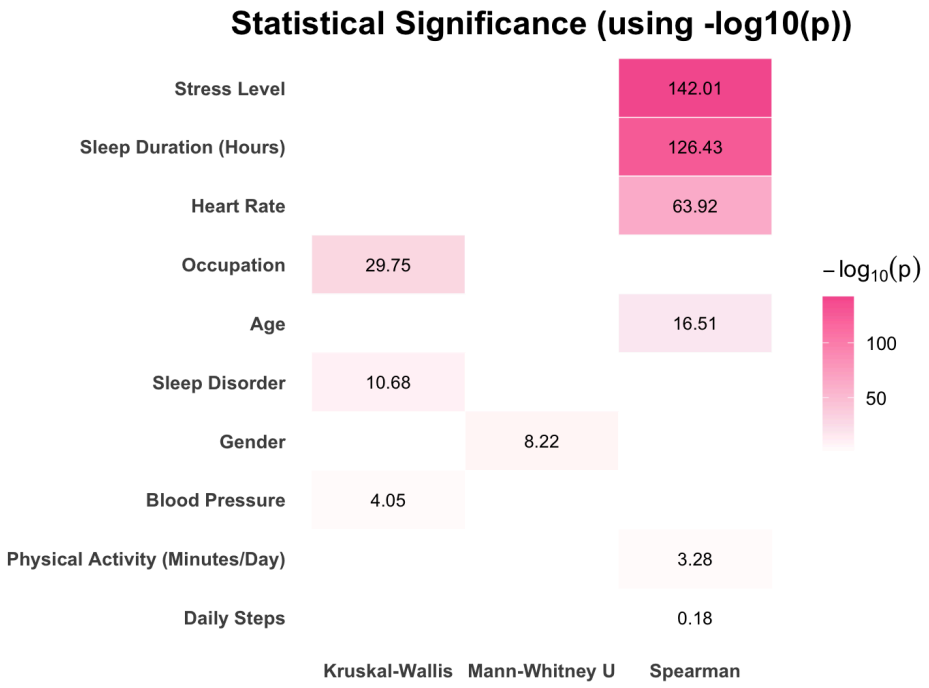


Figure 14. Heatmap showing the statistical significance of the variables in regards to Sleep Quality.

5.5. Linear Regression

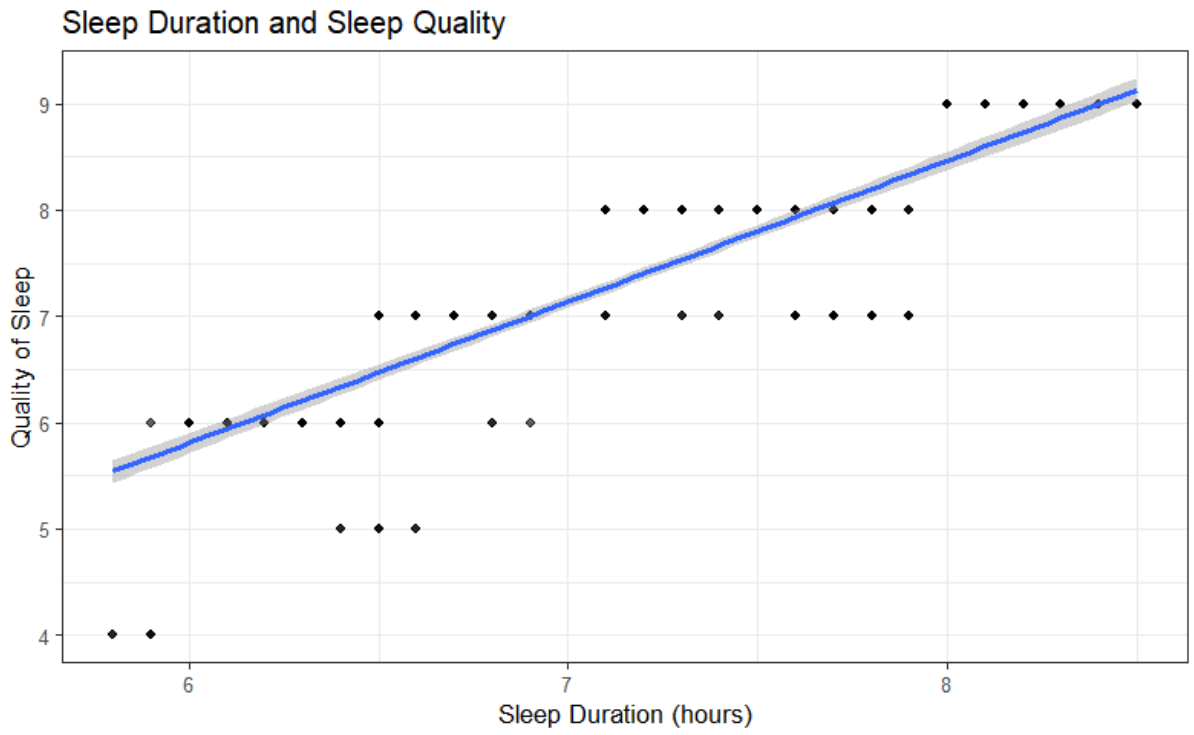


Figure 15. Scatter plot showing the relationship between sleep duration and sleep quality.

Figure 15 shows a positive linear relationship between sleep duration and sleep quality. Individuals with longer sleep durations generally reported higher sleep quality scores. The fitted regression line indicates that sleep quality tends to increase as sleep duration increases, suggesting that sleep duration is an important predictor of sleep quality.

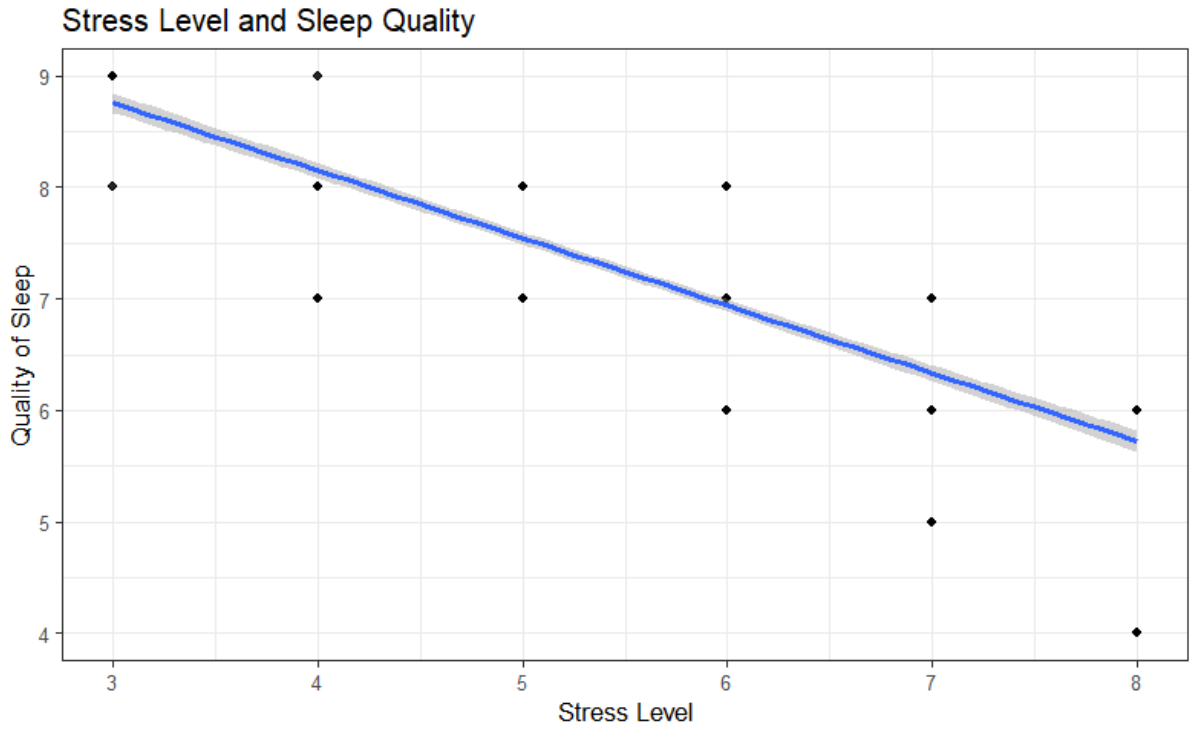


Figure 16. Scatter plot showing the relationship between stress level and sleep quality.

The figure above demonstrates a negative linear relationship between stress level and sleep quality. Higher stress levels were generally associated with lower sleep quality scores. This trend suggests that increasing stress is associated with poorer sleep quality.

The correlation analysis identified several strong linear associations between the numerical variables. Sleep quality was strongly positively correlated with sleep duration (r=0.88) and strongly negatively correlated with stress level (r=−0.90).

Daily steps and physical activity also showed a strong positive correlation (r=0.77), indicating that individuals with higher physical activity generally accumulated more daily steps. Heart rate exhibited a moderate negative correlation with sleep duration (r=−0.52) and sleep quality (r=−0.66). These findings provide an initial indication of the variables most strongly associated with sleep quality.

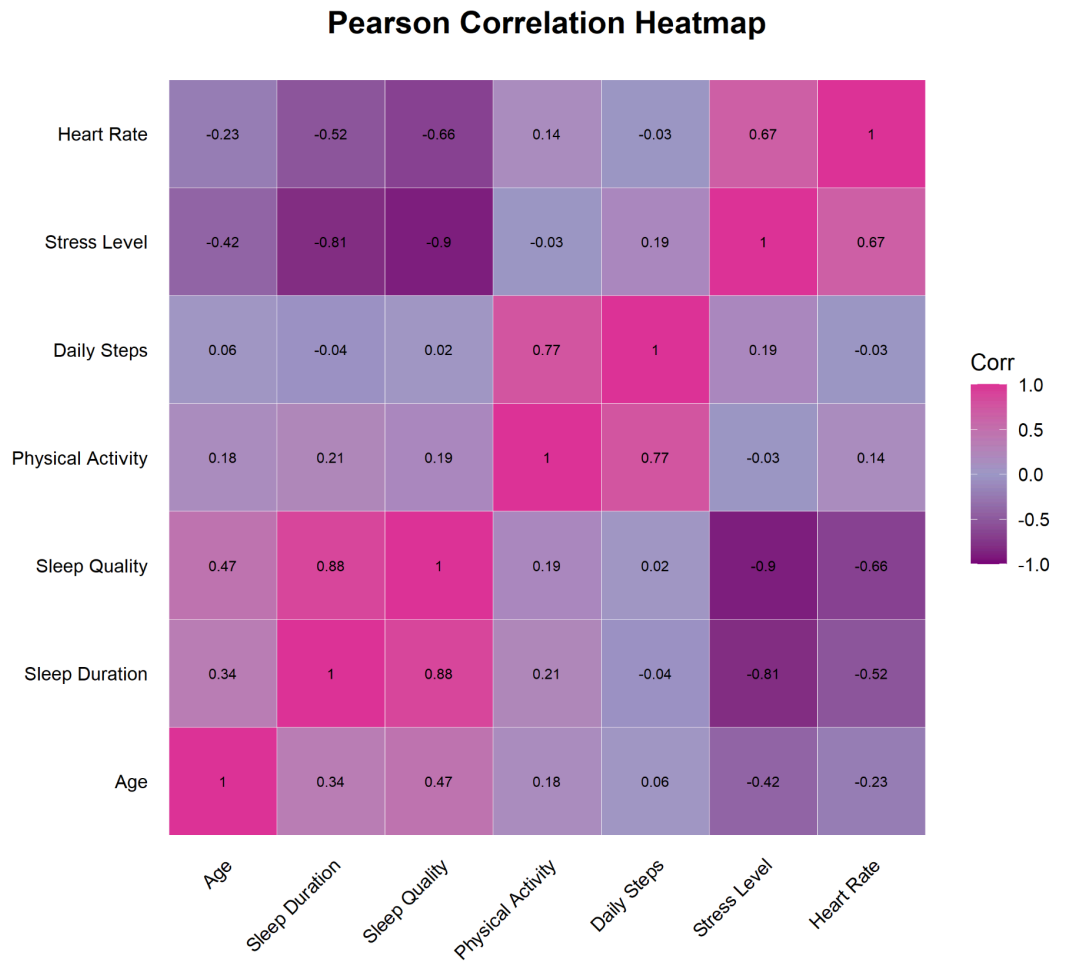


Figure 17. Pairwise correlations among the numerical predictors

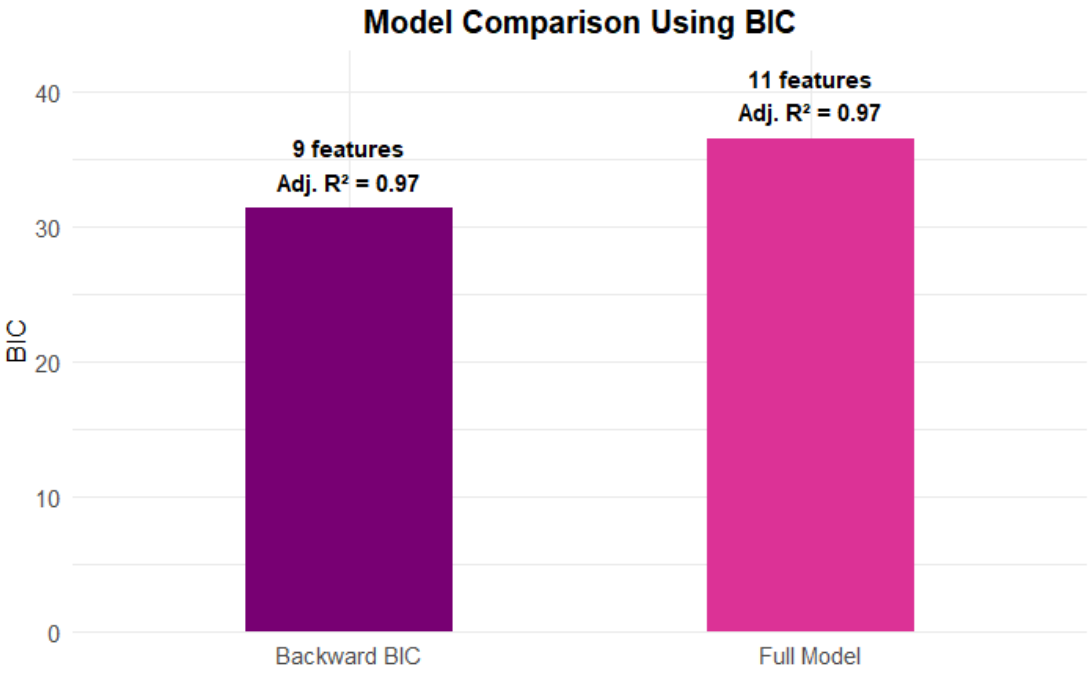


Figure 18. Comparison of the Full and Backward BIC-selected Models

The backward BIC model retained nine predictors compared with eleven predictors in the full model. Although the reduced model contained fewer predictors, it achieved an adjusted R^2 comparable to that of the full model while exhibiting a lower BIC value. This indicates that the BIC-selected model provides a more parsimonious representation of the data without compromising model fit.

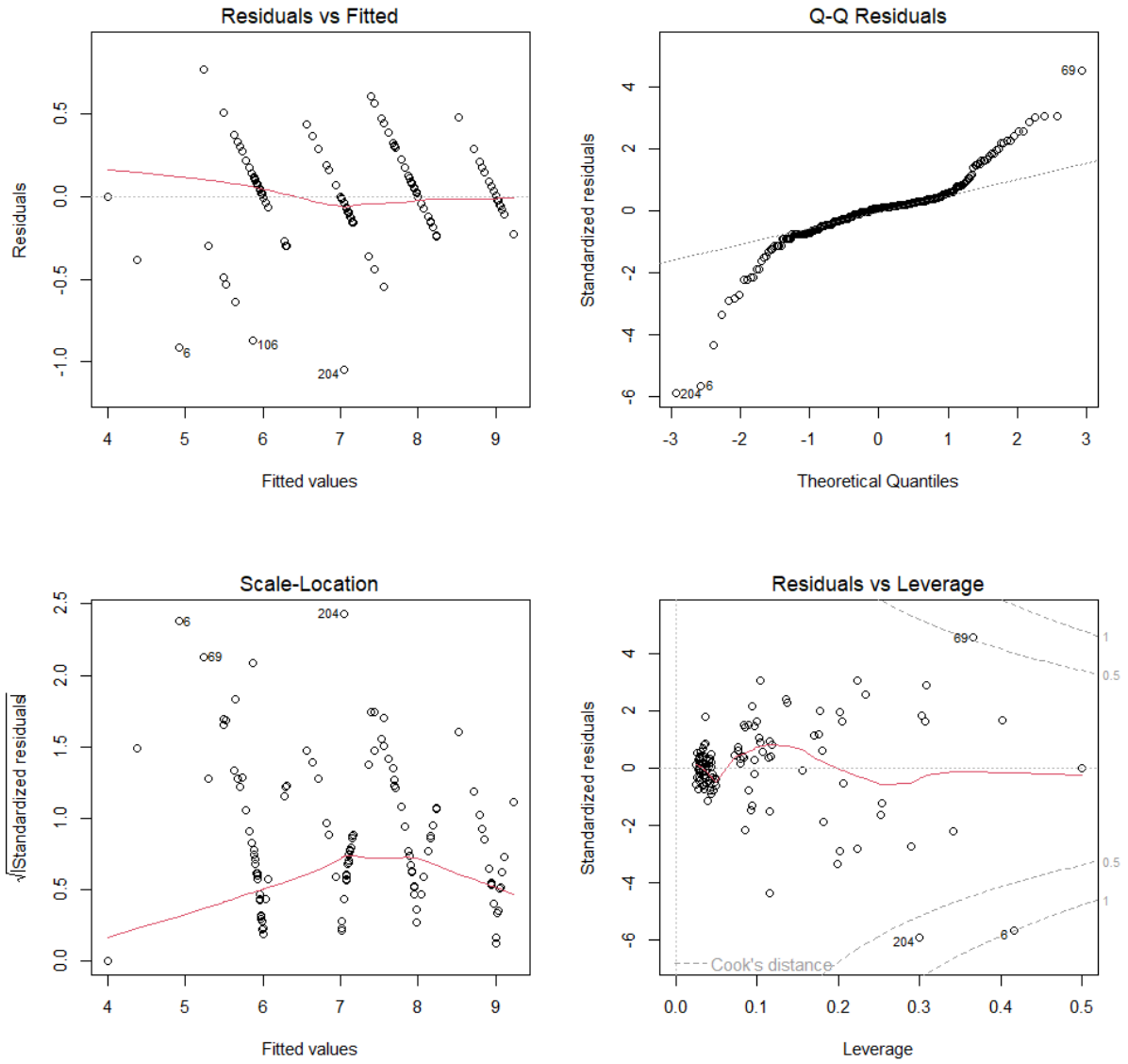


Figure 19. Standard diagnostic plots for the final regression model.

The diagnostic plots indicate that the assumptions of linear regression were reasonably satisfied. The residuals were randomly distributed around zero with no pronounced systematic pattern, suggesting that the assumption of linearity was largely met. The Normal Q–Q plot showed that most residuals followed the theoretical normal distribution, with slight deviations at the tails indicating the presence of a few extreme observations. The Scale–Location plot suggested relatively constant residual variance across the fitted values, although a small

increase in variability was observed in the middle range of fitted values. The Residuals versus Leverage plot did not reveal observations exceeding Cook's distance thresholds, indicating that no highly influential observations substantially affected the fitted model.

6. Discussion

Overall, the findings indicate that sleep quality is influenced by multiple factors, with stress level, sleep duration, occupation, and gender demonstrating the strongest relationships. In contrast, daily step count showed little association with perceived sleep quality.

The descriptive analyses revealed that most participants reported relatively good sleep quality, with scores primarily ranging between six and nine on the ten-point scale. Similarly, most participants reported sleeping between seven and eight hours per night, which aligns with the recommendations of Hyndych et al. (2025) where they reported that sleeping for approximately seven hours is the optimal amount for an ideal cognitive performance. The majority of participants also reported having no diagnosed sleep disorder, indicating that the dataset largely represents a healthy adult population rather than a clinical cohort.

The decision tree classifier selected the features of stress level and sleep duration as the most important predictors for sleep quality, the model showed an accuracy of 97 percent. For sleep disorder, accuracy was slightly lower by 6 percent, more impactful were occupation, age, and heart rate. This claim is further supported by prior findings that prove the correlation of psychological stress and low sleep quality (Baglioni et al., 2016). Cluster analysis then, separately, proved the existence of distinct clusters with specific feature combinations. Clusters with observations with lower and longer sleep duration had more observations with high sleep quality, in contrast, to ones with high stress levels which showed low or medium sleep quality and a higher probability for insomnia.

Among all variables examined, stress level emerged as the strongest predictor of sleep quality. The Spearman correlation showed a very strong negative relationship ($p = -0.91$), indicating that participants reporting higher stress levels consistently experienced poorer sleep quality. This finding is highly consistent with previous research by Zhang et al. (2024) which states that individuals reporting higher perceived stress also experienced significantly higher levels of anxiety, which was associated with lower perceived sleep quality.

Conversely, sleep duration demonstrated the strongest positive association with sleep quality ($p = 0.89$). Participants who slept longer generally reported better sleep quality, suggesting that adequate sleep duration remains one of the most important contributors to restorative sleep. While sleep duration and sleep quality represent distinct constructs, they are closely related, as insufficient sleep often reduces opportunities for completing normal sleep cycles and achieving restorative stages of sleep. Nevertheless, it is important to recognize that longer sleep duration does not

always guarantee high-quality sleep, as underlying sleep disorders or fragmented sleep may reduce sleep quality despite sufficient hours slept (Lallukka et al., 2018).

The multiple linear regression analysis further supported the findings obtained from the descriptive, correlation, decision tree, and cluster analyses. Backward feature selection using the Bayesian Information Criterion (BIC) reduced the model complexity by removing Physical Activity Level and Daily Steps, while retaining the remaining predictors. Despite using fewer predictors, the BIC-selected model achieved a lower BIC and predictive performance comparable to that of the full model. This suggests that the excluded variables contributed limited additional predictive information once the remaining predictors were included, resulting in a more parsimonious model without compromising model performance.

The final regression model explained a large proportion of the variability in sleep quality, as reflected by the high adjusted R-squared. Furthermore, standard diagnostic analyses indicated that the assumptions of linear regression were reasonably satisfied. The residual plots showed no substantial violations of linearity, normality, or homoscedasticity. In addition, the adjusted Generalized Variance Inflation Factor (GVIF) values for all predictors were below the commonly accepted threshold of 5, indicating low multicollinearity. This suggests that the regression coefficient estimates are stable and that the retained predictors contribute largely independent information to the model.

Limitations: The dataset used in this study is synthetic and relatively small (374 observations), increasing the likelihood that the identified patterns are specific to the simulated data rather than representative of real-world populations. Furthermore, no hyperparameter tuning, cross-validation, or external validation was performed, which may have resulted in optimistic estimates of predictive performance and increased the risk of overfitting. Finally, while cluster analysis was useful for identifying groups of participants with similar characteristics, more advanced machine learning methods, such as Random Forests, Gradient Boosting, or XGBoost, could provide additional insights and should be considered in future studies.

7. Conclusion

This study analysed how lifestyle and health-related factors influence sleep quality using descriptive statistics, machine learning techniques, non-parametric statistical tests, and linear regression. Across different analytical approaches, stress level and sleep duration consistently appeared as the variables with the strongest association with sleep quality, while occupation also demonstrated an important influence, particularly in the prediction of sleep disorders and differences in sleep quality between occupational groups.

The descriptive analyses performed, showed that most participants reported moderate to good sleep quality and sleep durations close to the recommended seven to eight hours per night. Decision tree models achieved high predictive performance, identifying stress level and sleep duration as the primary determinants of sleep quality, whereas occupation, age, and heart rate were the most relevant

predictors of sleep disorders. Similarly, the cluster analysis identified distinct participant groups characterized by different combinations of lifestyle and health-related factors, further highlighting the relationship between higher stress, shorter sleep duration, and poorer sleep outcomes.

The non-parametric statistical analyses supported these findings. Spearman correlation demonstrated a strong positive association between sleep duration and sleep quality and a strong negative association between stress level and sleep quality. Significant differences were also observed across occupation groups, gender, sleep disorder status, blood pressure categories, and physical activity, whereas daily steps showed little association with sleep quality. Linear regression further confirmed that sleep duration and stress level were among the strongest contributors to the prediction of sleep quality. Multiple linear regression further confirmed these findings, with the BIC-selected model explaining a large proportion of the variability in sleep quality while maintaining predictive performance using fewer predictors than the full model. This demonstrates that a more parsimonious model can effectively explain sleep quality without compromising model performance.

Although the study provides consistent evidence that stress management, adequate sleep duration, and occupational factors are important contributors to sleep quality, several limitations should be considered. The dataset was synthetic, relatively small, and not necessarily representative of the general population. Furthermore, the absence of cross-validation, and external validation may have resulted in optimistic estimates of model performance.

Overall, despite these limitations, the findings demonstrate that combining statistical methods with machine learning provides complementary insights into factors associated with sleep quality. Future research should validate these findings using larger, real-world datasets and more advanced predictive models to improve generalizability and better understand the complex interactions between lifestyle, health status, and sleep.

8. References

Baglioni, C., Battagliese, G., Feige, B., Spiegelhalder, K., Nissen, C., Voderholzer, U., Lombardo, C., & Riemann, D. (2016). Insomnia as a predictor of depression: A meta-analytic evaluation of longitudinal epidemiological studies. Journal of Affective Disorders, 186, 47–55. https://doi.org/10.1016/j.jad.2015.06.008

Baranwal, N., Yu, P. K., & Siegel, N. S. (2023). Sleep physiology, pathophysiology, and sleep hygiene. Progress in cardiovascular diseases, 77, 59–69. https://doi.org/10.1016/j.pcad.2023.02.005

Hyndych, A., El-Abassi, R., & Mader, E. C. (2025). The Role of Sleep and the Effects of Sleep Loss on Cognitive, Affective, and Behavioral Processes. Cureus, 17(5). https://doi.org/10.7759/cureus.84232

Jaspan, V. N., Greenberg, G. S., Parihar, S., Park, C. M., Somers, V. K., Shapiro, M. D., Lavie, C. J., Virani, S. S., & Slipczuk, L. (2024). The Role of Sleep in Cardiovascular Disease. Current atherosclerosis reports, 26(7), 249–262. https://doi.org/10.1007/s11883-024-01207-5

Lallukka, T., Sivertsen, B., Kronholm, E., Bin, Y. S., Øverland, S., & Glozier, N. (2018). Association of sleep duration and sleep quality with the physical, social, and emotional functioning among Australian adults. Sleep Health, 4(2), 194–200. https://doi.org/10.1016/j.sleh.2017.11.006

Mayo Clinic. (n.d). “Blood pressure chart: What your reading means”. https://www.mayoclinic.org/diseases-conditions/high-blood-pressure/in-depth/blood-pressure/art-20050982

Oros, M., Franck Soyez, Moldovan, A.-D., Aliuș Ruxandra Oana, Voicu, B., & Florin Mihaltan. (2025). Sleep and modern life: A population-based study. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-13405-5

Tharmalingam, L. (2023). Sleep Health and Lifestyle Dataset. Kaggle. https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset

Tompa, R. (2025). How Sleep Affects Mental Health (and Vice Versa): What the Science Says. News Center. https://med.stanford.edu/news/insights/2025/08/sleep-mental-health-connection-what-science-says.html

Zhang, J., Xiang, S., Li, X., Tang, Y., & Hu, Q. (2024). The impact of stress on sleep quality: A mediation analysis based on longitudinal data. *Frontiers in Psychology*, *15*. https://doi.org/10.3389/fpsyg.2024.1431234